



Erratum

Erratum to “A priori and a posteriori error estimates for the Deep Ritz method applied to the Laplace and Stokes problem” [J. Comput. Appl. Math. 421 (2023) 114845]

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A B S T R A C T

The proof of Lemma 1 contains careless errors and is based on a faulty approach. We provide a correction here.

1. Correction to the proof of Lemma 1

The original proof given to Lemma 1 is based on a faulty approach. The statement of the Lemma itself is valid and in the following we give a corrected proof. The Lemma itself is identical to the version provided in the original manuscript and the correction fits seamlessly into the manuscript. We have found a proof for a corresponding lemma in the literature, see [1, Theorem 4]. Nevertheless, we present a proof that is in parts inspired by [1], but which is formulated in the context of standard variational analysis.

Lemma 1 (Model Error). Let $f \in L^2(\Omega)$, $\lambda \in \mathbb{R}_+$ and Ω be such that the solutions $u \in H_0^1(\Omega)$ and $u_\lambda \in H^1(\Omega)$ to

$$(\nabla u, \nabla \phi) = (f, \phi), \quad (\nabla u_\lambda, \nabla \phi_\lambda) + \lambda(u_\lambda, \phi_\lambda) = (f, \phi) \quad (1)$$

for $\phi \in H_0^1(\Omega)$ and $\phi_\lambda \in H^1(\omega)$ satisfy $\|u\|_{H^2(\Omega)} + \|u_\lambda\|_{H^2(\Omega)} \leq c_s \|f\|$. It holds

$$\|\nabla(u - u_\lambda)\| \leq \frac{c}{\lambda} \|f\|,$$

where $c > 0$ depends on the domain Ω only.

Proof. For u and u_λ it holds

$$\begin{aligned} -\Delta u &= f \text{ in } \Omega, & u &= 0 \text{ on } \partial\Omega, \\ -\Delta u_\lambda &= f \text{ in } \Omega, & \lambda u_\lambda + \partial_n u_\lambda &= 0 \text{ on } \partial\Omega. \end{aligned}$$

We define $\psi \in H^1(\Omega)$ as weak solution to

$$-\Delta \psi = 0 \text{ in } \Omega, \quad \psi = \partial_n u \text{ on } \partial\Omega.$$

It holds $\|\nabla \psi\| \leq c \|f\|$ as $\|\partial_n u\|_{H^{\frac{1}{2}}(\partial\Omega)} \leq c \|u\|_{H^2(\Omega)} \leq c \|f\|$. Similar to [1, Theorem 4] we define

$$w = u - \frac{1}{\lambda} \psi. \quad (2)$$

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For the error it holds

$$\|\nabla(u - u_\lambda)\|^2 = (\nabla(u - u_\lambda), \nabla(u - w)) + (\nabla(u - u_\lambda), \nabla(w - u_\lambda)), \quad (3)$$

and, based on (2), the first term is estimated as

$$\begin{aligned} (\nabla(u - u_\lambda), \nabla(u - w)) &\leq \frac{1}{2} \|\nabla(u - u_\lambda)\|^2 + \frac{1}{2} \|\nabla(u - w)\|^2 \\ &= \frac{1}{2} \|\nabla(u - u_\lambda)\|^2 + \frac{1}{2\lambda^2} \|\nabla\psi\|^2 \leq \frac{1}{2} \|\nabla(u - u_\lambda)\|^2 + \frac{c}{\lambda^2} \|f\|^2. \end{aligned} \quad (4)$$

The first term will be absorbed in the left hand side of (3). The second term of (3) gets

$$(\nabla(u - u_\lambda), \nabla(w - u_\lambda)) = (-\Delta(u - u_\lambda), w - u_\lambda) + \langle \partial_n(u - u_\lambda), w - u_\lambda \rangle_{\partial\Omega}, \quad (5)$$

where the first part is zero. Using (2) as well as $\lambda u_\lambda + \partial_n u_\lambda = 0$ it holds

$$\begin{aligned} (\nabla(u - u_\lambda), \nabla(w - u_\lambda)) &= \langle \partial_n(u - u_\lambda), u - \frac{1}{\lambda} \partial_n u - u_\lambda \rangle_{\partial\Omega} \\ &= -\frac{1}{\lambda} \|\partial_n u + \lambda u_\lambda\|_{L^2(\partial\Omega)}^2 \leq 0 \end{aligned} \quad (6)$$

Combining the above gives the error estimate. $\square$

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Data availability

No data was used for the research described in the article.

References

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